MLOPS Assignment 2

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1. Which run had the lowest latency, and why might that be?

- run_batch8 had the lowest latency, at 0.120 s.
- run_batch4 took 0.133 s, and run_batch1 was the slowest at 0.226 s.
- With batch size 8, the model processes 8 sentences together in one forward pass, so the 10 sentences need only 2 model calls.
- Fewer calls means less overhead per call, so the total time is lower.
- run_batch1 made 10 separate calls, and since it ran first it also paid a one-time warm-up cost.
- So latency decreases as batch size increases.

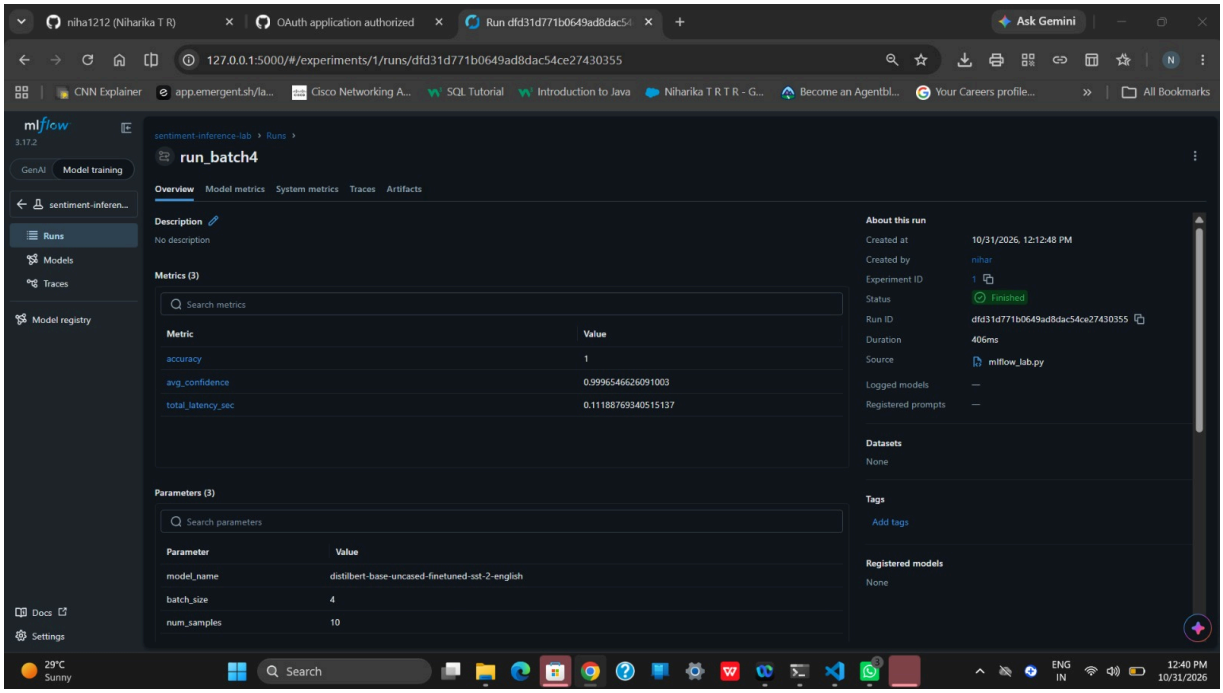
2. Did accuracy change between runs? Why or why not?

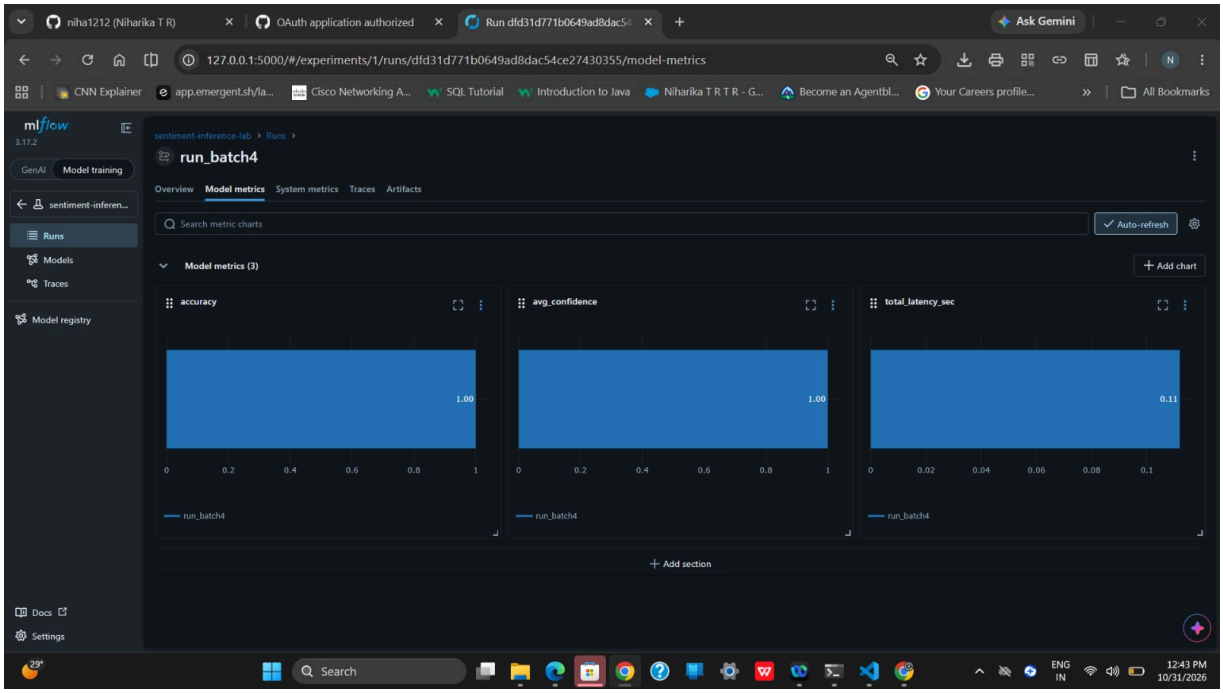
- No, accuracy stayed at 1.0 (100%) in all three runs.
  Every run used the same pre-trained DistilBERT model with the same weights.
- All runs were tested on the same 10 sentences.
- Inference is deterministic, meaning the same input always gives the same output.
- Batch size only changes how fast the sentences are processed, not the predictions.
- Average confidence was also the same (0.99965) in all runs.

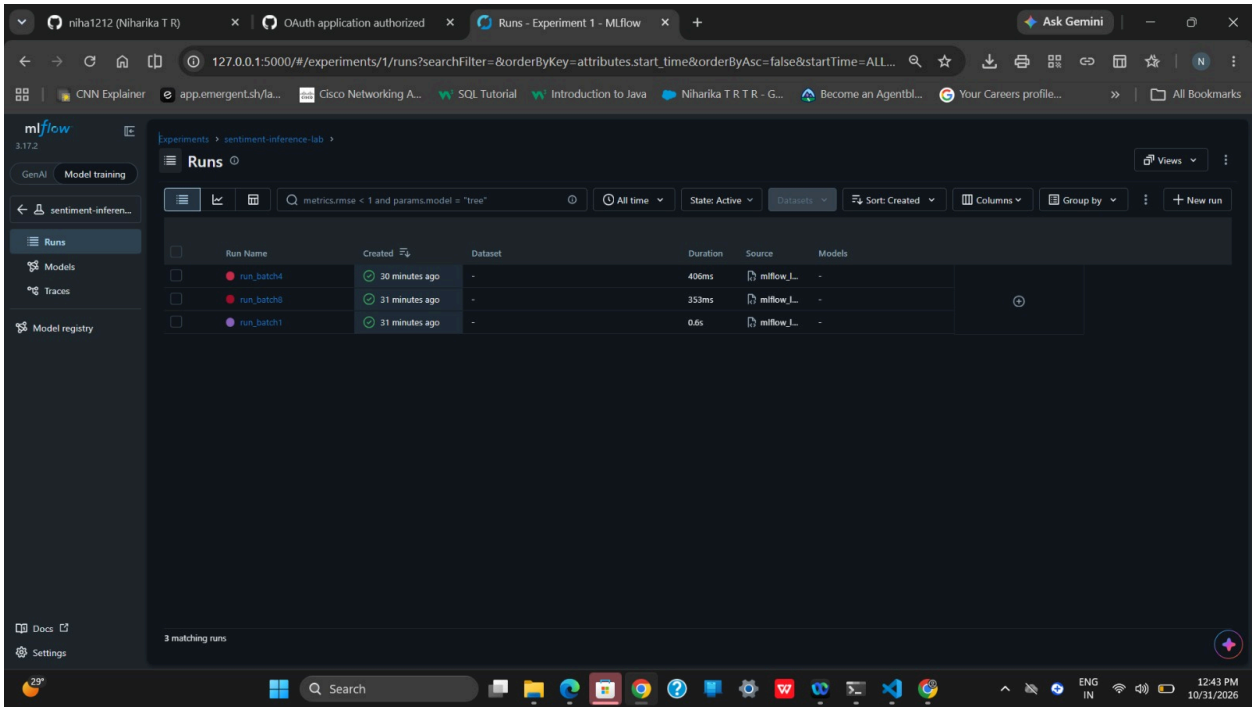
3. Name two benefits of using MLflow here instead of printing results in the terminal.

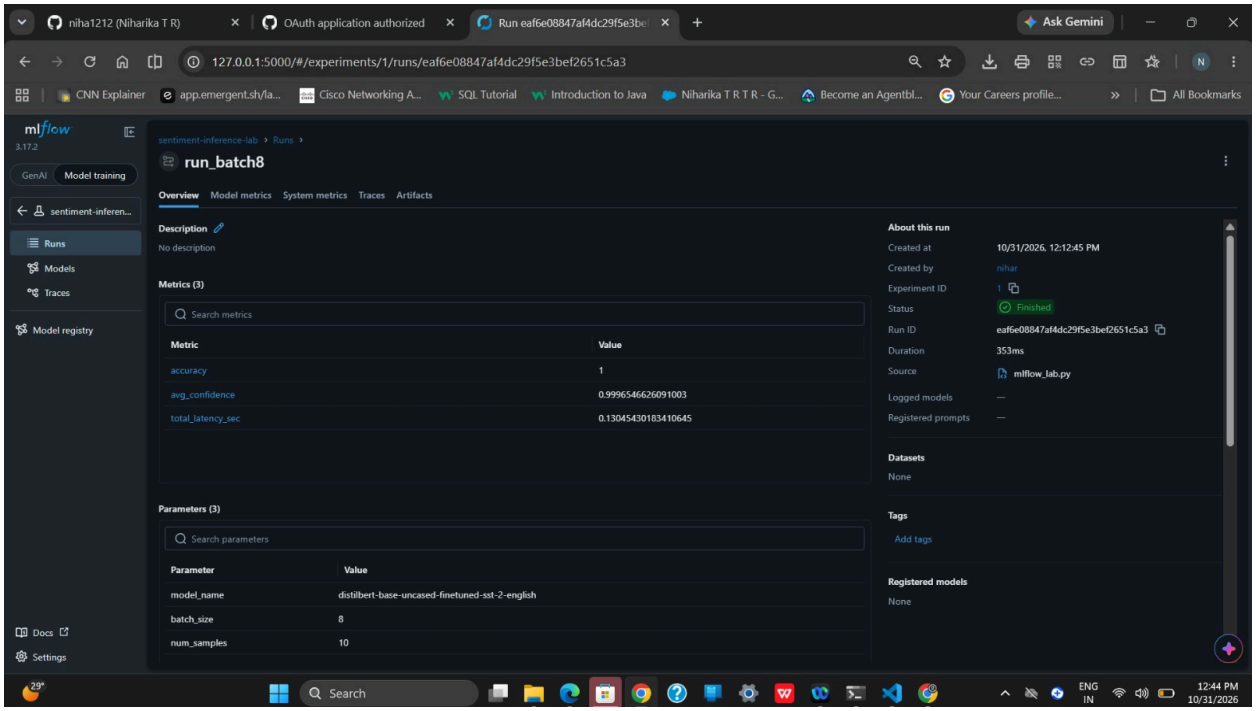
- Reproducibility and record-keeping: the parameters, metrics and artifacts (predictions.csv) of every run are saved permanently. Terminal output is lost once the window closes.
- Easy comparison: runs can be viewed and compared side by side in the MLflow UI, with tables and charts, instead of reading through printed lines.

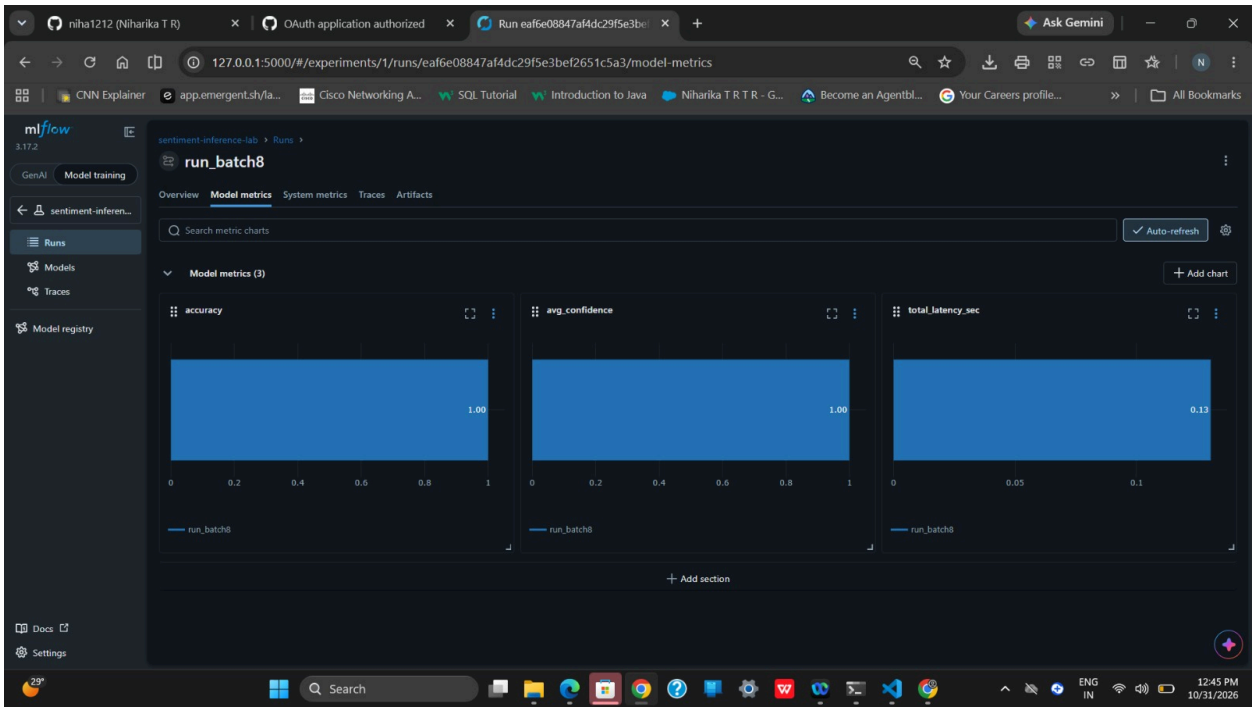
Screenshot of the MLflow runs table:

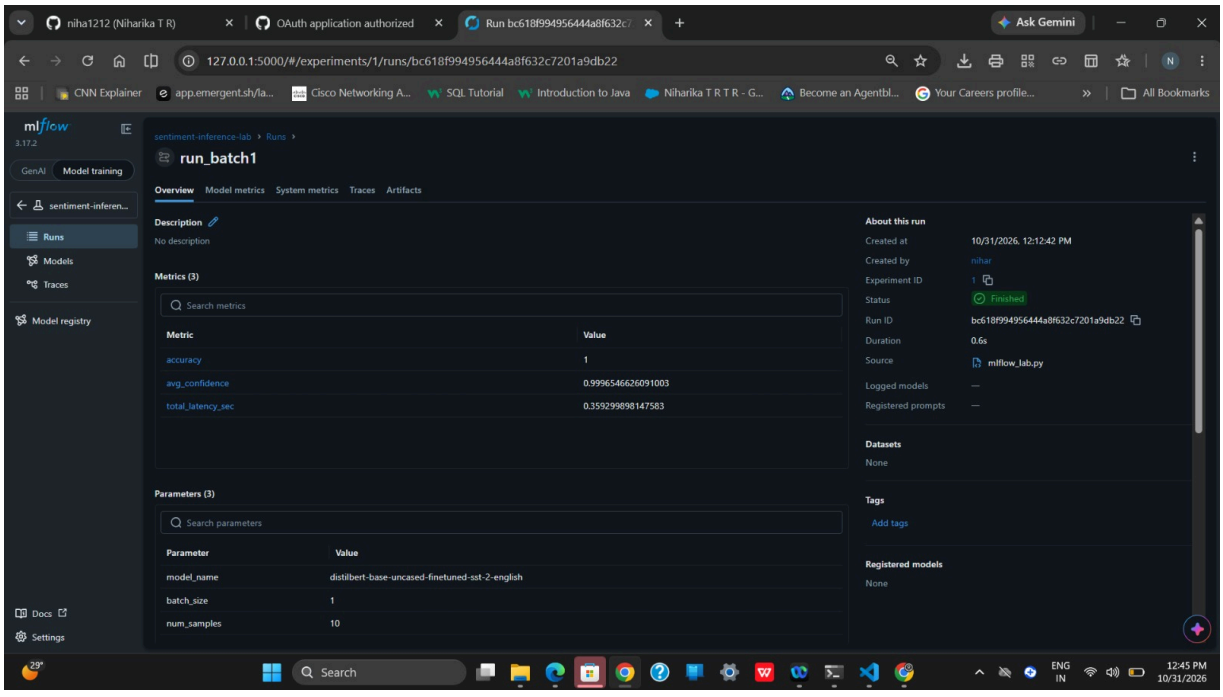


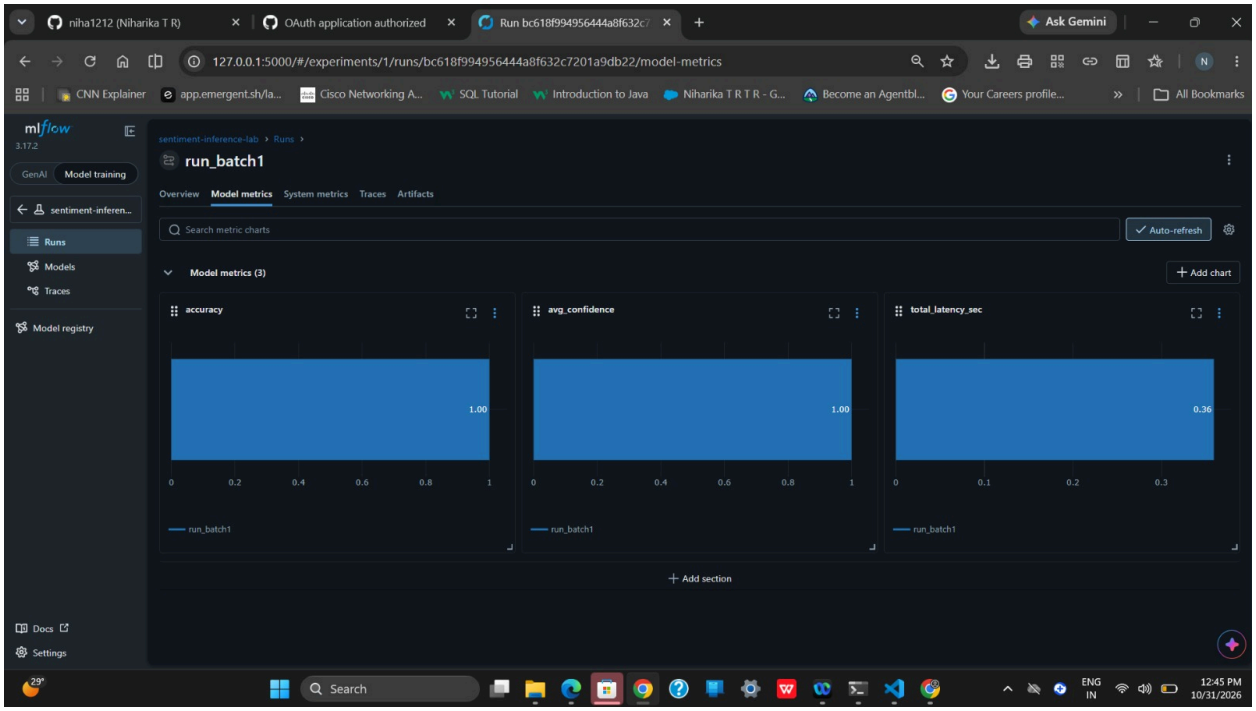



niha1212 (Niharika T R)
OAuth application authorized
Runs - Experiment 1 - MLflow
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs?searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL...
CNN Explainer
app.emergent.sh/la...
Cisco Networking A...
SQL Tutorial
Introduction to Java
Niharika T R T R - G...
Become an Agentbl...
Your Careers profile...
All Bookmarks
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
Experiments › sentiment-inference-lab ›
Runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
State: Active
Datasets
Sort: Created
Columns
Group by
New run
Run Name
Created
Dataset
Duration
Source
Models
run_batch4
30 minutes ago
406ms
mlflow_l...
run_batch8
31 minutes ago
353ms
mlflow_l...
run_batch1
31 minutes ago
0.6s
mlflow_l...
3 matching runs
Search
ENG
IN
12:43 PM
10/31/2026

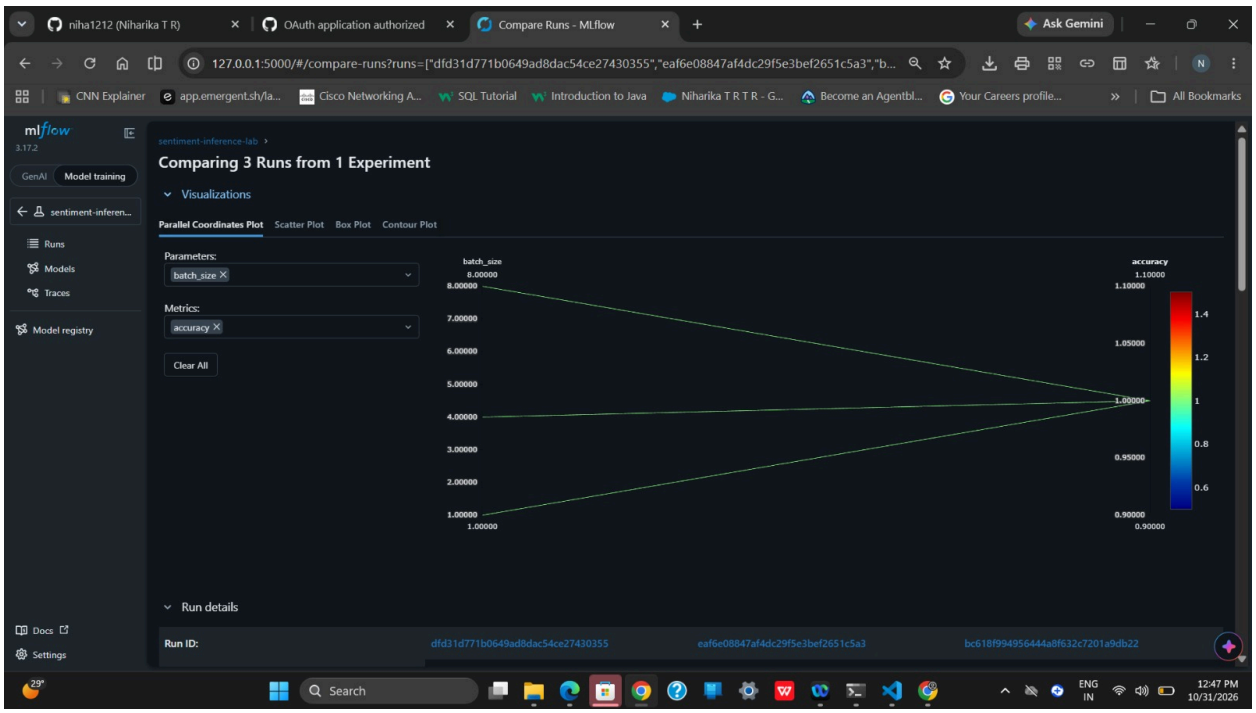

niha1212 (Niharika T R)
OAuth application authorized
Run eaf6e08847af4dc29f5e3bef
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/eaf6e08847af4dc29f5e3bef2651c5a3
CNN Explainer
app.emergent.sh/la...
Cisco Networking A...
SQL Tutorial
Introduction to Java
Niharika T R T R - G...
Become an Agentbl...
Your Careers profile...
All Bookmarks
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.13045430183410645
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/31/2026, 12:12:45 PM
Created by
nihar
Experiment ID
1
Status
Finished
Run ID
eaf6e08847af4dc29f5e3bef2651c5a3
Duration
353ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Search
ENG
IN
12:44 PM
10/31/2026


niha1212 (Niharika T R)
OAuth application authorized
Run eaf6e08847af4dc29f5e3bef
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/eaf6e08847af4dc29f5e3bef2651c5a3/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.13
run_batch8
Add section
Docs
Settings
Search
ENG
IN
12:45 PM
10/31/2026

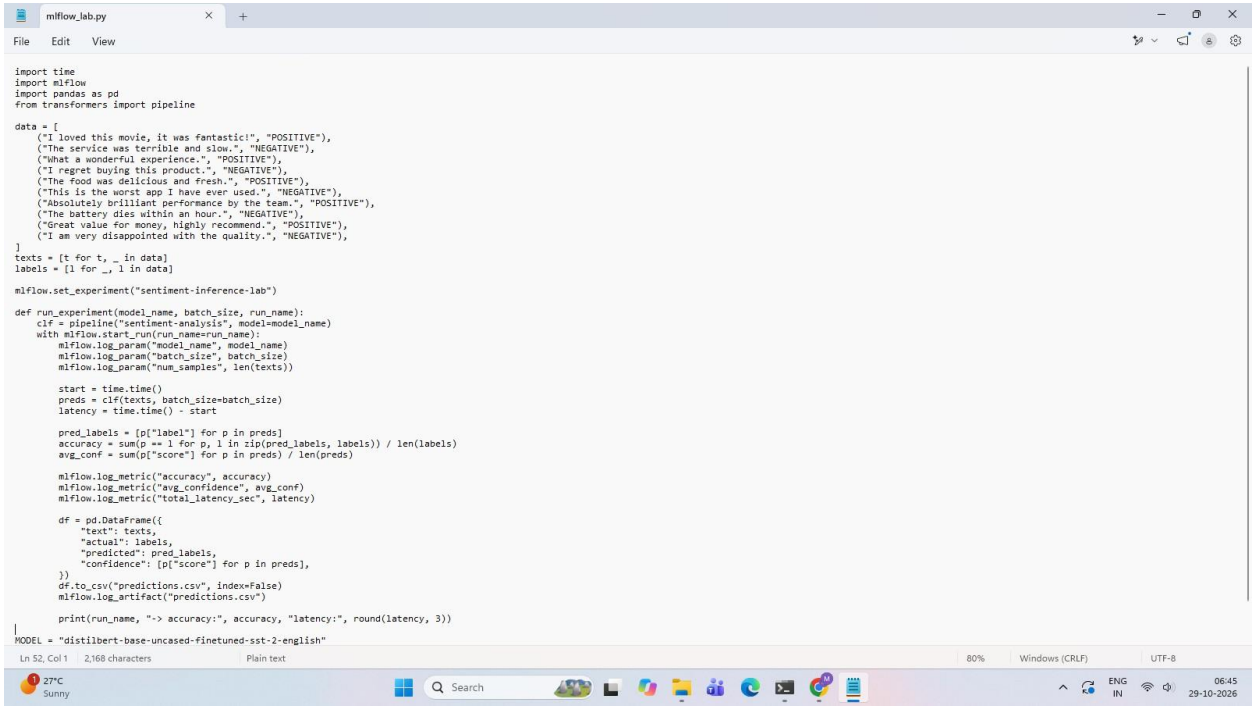

niha1212 (Niharika T R)
OAuth application authorized
Run bc618f994956444a8f632c7
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/bc618f994956444a8f632c7201a9db22
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.359299898147583
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/31/2026, 12:12:42 PM
Created by
nihar
Experiment ID
1
Status
Finished
Run ID
bc618f994956444a8f632c7201a9db22
Duration
0.6s
Source
mlflow_lab.py
Logged models
Registered prompts
Datasets
None
Tags
Add tags
Registered models
None
Docs
Settings
Search
ENG
IN
12:45 PM
10/31/2026



Screenshot of the Compare view:



mlflow_lab.py file:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

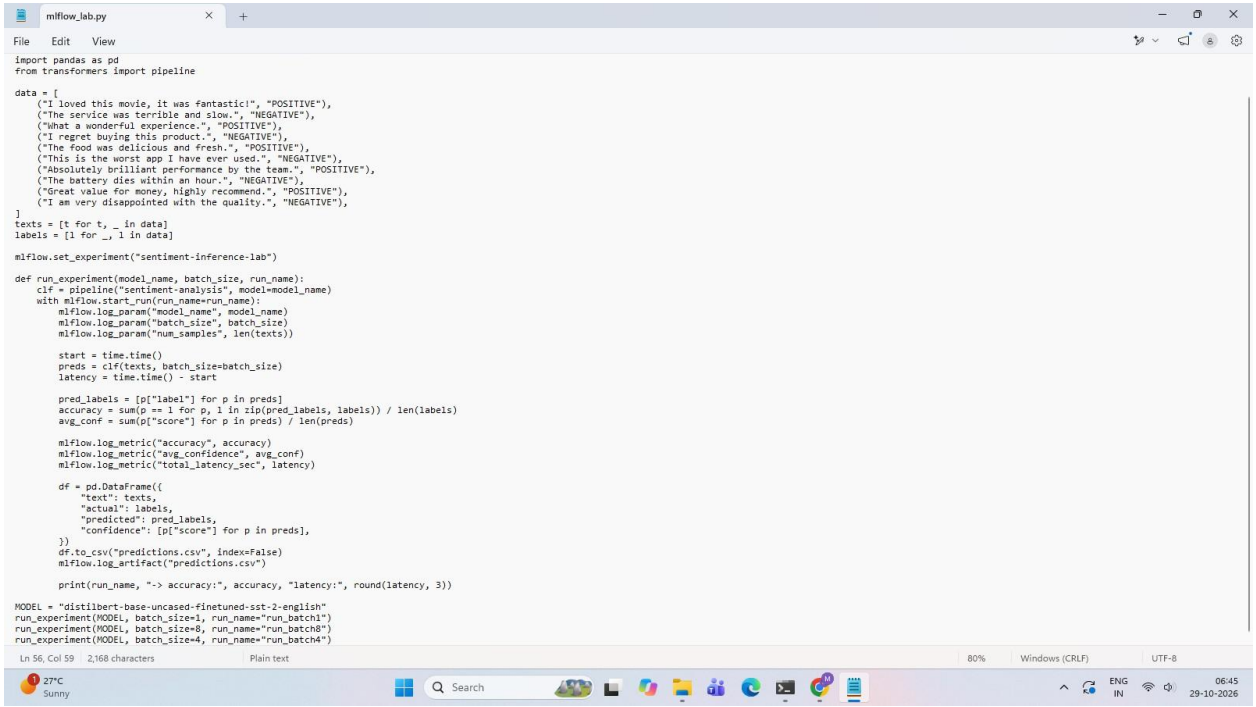
        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
```



```
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

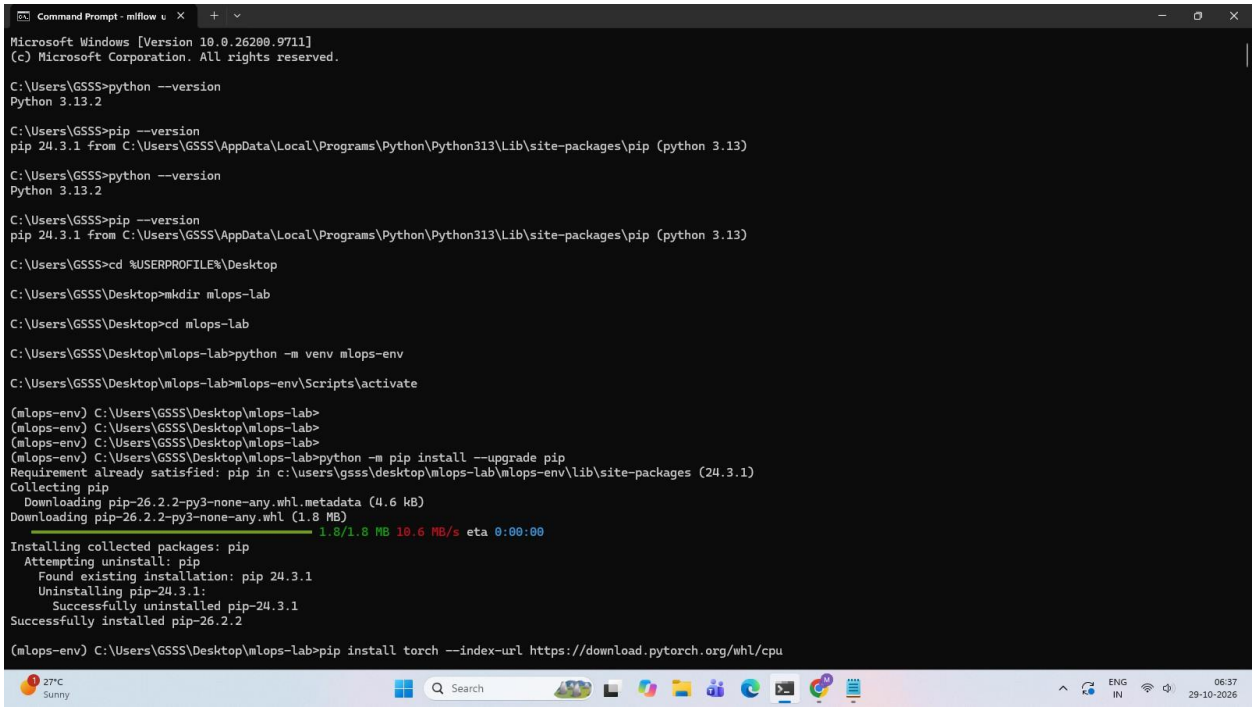
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

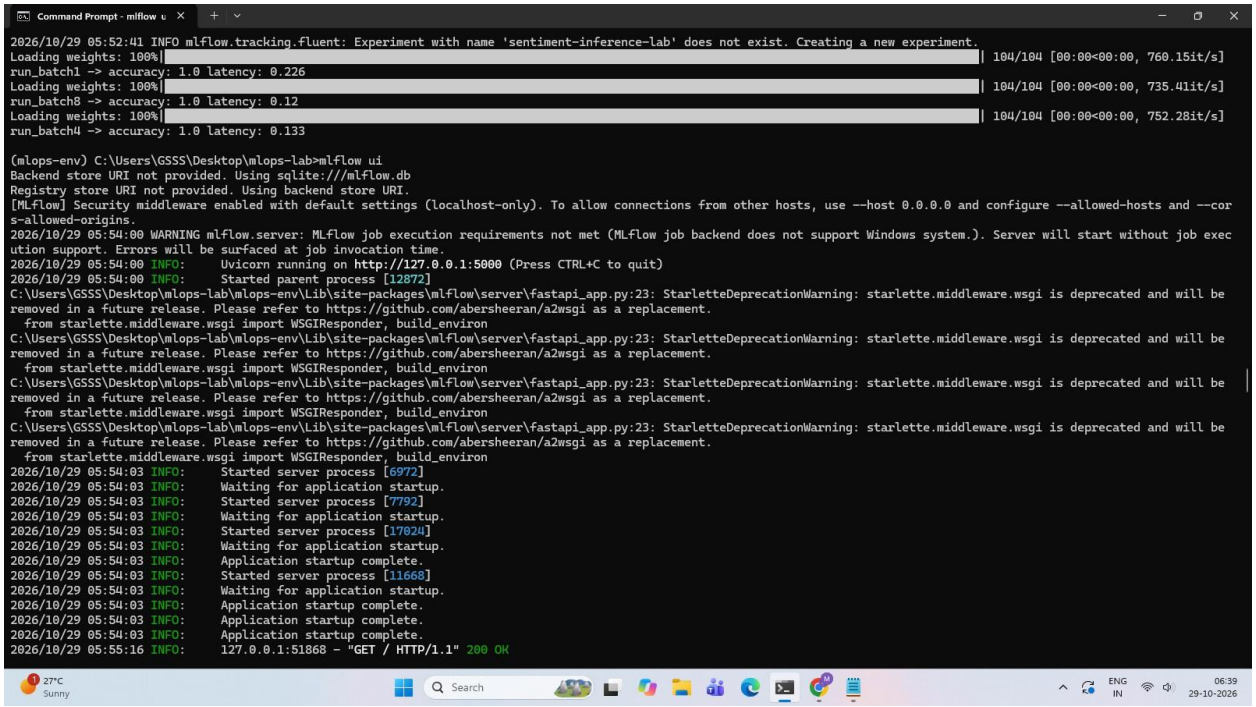
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Execution:


Command Prompt - mlflow u
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>cd %USERPROFILE%\Desktop
C:\Users\GSSS\Desktop>mkdir mlops-lab
C:\Users\GSSS\Desktop>cd mlops-lab
C:\Users\GSSS\Desktop\mlops-lab>python -m venv mlops-env
C:\Users\GSSS\Desktop\mlops-lab>mlops-env\Scripts\activate
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\gsss\desktop\mlops-lab\mlops-env\lib\site-packages (24.3.1)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 10.6 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 24.3.1
Uninstalling pip-24.3.1:
Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.2
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>pip install torch --index-url https://download.pytorch.org/whl/cpu
27°C
Sunny
Search
ENG
IN
06:37
29-10-2026


Command Prompt - mlflow u
2026/10/29 05:52:41 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 760.15it/s]
run_batch1 -> accuracy: 1.0 latency: 0.226
Loading weights: 100%| 104/104 [00:00<00:00, 735.41it/s]
run_batch8 -> accuracy: 1.0 latency: 0.12
Loading weights: 100%| 104/104 [00:00<00:00, 752.28it/s]
run_batch4 -> accuracy: 1.0 latency: 0.133
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 05:54:00 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 05:54:00 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 05:54:00 INFO: Started parent process [12872]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 05:54:03 INFO: Started server process [6972]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [7792]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [17024]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Started server process [11668]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:55:16 INFO: 127.0.0.1:51868 - "GET / HTTP/1.1" 200 OK
27°C
Sunny
Search
ENG
IN
06:39
29-10-2026